7.2 Error in Methods for Stochastic Differential Equations

We briefly present some standard material on the development of numerical schemes for SDEs. The interested reader is pointed to the standard textbooks [200, 270] for a more thorough discussion of foundational concepts. Let us begin with a discussion of the first order stochastic differential equation

$$\mathrm{dX} = a(\mathrm{X})\mathrm{d}t + b(\mathrm{X})\mathrm{dW},$$

where W defines a standard scalar Wiener process. The natural extension of the Euler method for SDEs like this one takes the form

$$\mathrm{X}_{n+1} = \mathrm{X}_n + ha(\mathrm{X}_n) + b(\mathrm{X}_n)\sqrt{h}\mathrm{R}_n,$$

where R_n is a normal random variable with zero mean and unit variance. This method is referred to as the Euler-Maruyama method.

Note that from the definition of the SDE, we may write

$$\mathrm{X}(t+h) = \mathrm{X}(t) + \int_t^{t+h} a(\mathrm{X}(s))\mathrm{d}s + \int_t^{t+h} b(\mathrm{X}(s))\mathrm{dW}(s).$$

Just as in the deterministic case, we may build a numerical method by approximating these integrals over a small timestep h. However, when designing a numerical scheme it is important to keep in mind the goal of the discretization itself. For example, as we have considered Langevin dynamics as a tool for studying the canonical distribution, do we always need methods to give us good approximations of $\mathrm{X}(t+h)$? If we seek to compute averages with respect to the canonical distribution, then the answer is no: we simply require that the average over the discretized trajectory is accurate.

The numerical analysis of stochastic differential equations is traditionally based on the concepts of *weak* and *strong* accuracy. Let a numerical method be given for solving an SDE in the form of a discrete stochastic process $\mathrm{X}_{n+1} = \Phi(\mathrm{X}_n, h)$ for $n = 0, 1, \ldots, \nu - 1$, where $\nu h = \tau$ is fixed. We denote the stochastic solution of the SDE by $\mathrm{X}(t)$. In the case of strong accuracy, our measure of the global error is the quantity [200]

$$\mathrm{Err}^{\mathrm{strong}}_{h,\tau;\alpha} = \max_{n=0,1,2,\ldots,\nu} \left[\mathbb{E}\left|\mathrm{X}_n - \mathrm{X}(nh)\right|^\alpha\right]^{1/\alpha}.$$

For $\alpha = 1$ this implies convergence of expectation. The more natural choice from the probabilistic point of view is $\alpha = 2$ (mean square convergence). Having settled on a choice of α, we say that a method has strong order r if $\mathrm{Err}^{\mathrm{strong}}_{h,\tau} < C(\tau)h^r$ for some coefficient $C(\tau)$ depending on the time interval.